「Lemma. Let $g:[c,d]\to[a,b]$ be a reparametrization of a curve $\alpha:[a,b]\to\mathbb{R}^3$.
Then, the length of α is equal to the length of $\beta=\alpha\circ g$.

Proof.

$$\text{length}(\beta)=\int_c^d\left|\frac{d\beta}{dr}\right|dr=\int_c^d\left|\frac{d\alpha}{dg}\cdot\frac{dg}{dr}\right|dr=\int_c^d\left|\frac{d\alpha}{dg}\right|\cdot\left|\frac{dg}{dr}\right|dr$$

WLOG, $dg/dr>0$, then

$$=\int_c^d\left|\frac{d\alpha}{dg}\right|\frac{dg}{dr}\,dr=\int_a^b\left|\frac{d\alpha}{dt}\right|dt=\text{length}(\alpha).\ \blacksquare$$

If $\beta(s)$ is parametrized by arc length, then

$$|\beta'(t)|=\left|\frac{d\beta}{ds}\cdot\frac{ds}{dt}\right|=\left|\frac{d\beta}{ds}\right|\cdot\frac{ds}{dt}=1$$

$$\Rightarrow\left|\frac{d\beta}{ds}\right|=\frac{dt}{ds}=\frac{1}{ds}$$

$$\Rightarrow s=\int_0^s\left|\frac{d\beta}{du}\right|du,$$